

Discovery Piscine - AI/ML

Module 0 - Expert System

Summary: Regulate your indoor temperature as it fluctuates outside.

Version: 1.01

Contents

I	**A word about this Discovery Piscine**	**2**
II	**General instructions**	**3**
III	**Preamble**	**4**
IV	**Exercise 0: Environment**	**5**
V	**Exercise 1: Expert System**	**7**
VI	**Submission and peer-evaluation**	**9**

Chapter I

A word about this Discovery Piscine

Welcome!

This is a Discovery Piscine focused on artificial intelligence, from a coding perspective. You will discover various forms of artificial intelligence, both old and new, simple and complex. Many aspects of AI have actually been part of our daily lives for a long time. Of course, you will also experience 42's unique educational model: the peer-learning.

Basic Python knowledge is required to complete this Discovery Piscine.

`Regarding the peer-learning approach`:
You will encounter small challenges, projects and exercises.
At 42, there are no lectures, no courses, and no teachers.
Your task is to search for information, and to test it in order to verify that it is accurate, relevant, useful and not obsolete.
You usually can't do this alone. You will talk, exchange, and debate with your neighbors, to get new ideas and finally reach the goal.
There is no unique solution, and as long as your code produce the expected behavior and results, it's ok!

Peer-evaluation is a key moment to discover other ways of doing things, and exchange again on the project with different points of views.
You may realise that you missed something important,and need to do the same project again.

By the end of this Piscine, you will not have achieved the same progress as the other participants, you will not have learned the same knowledge. And that's totally ok!
This Discovery Piscine is at the same time a collective and a personal experience, and everyone will benefit from what this experience.

Good luck to all, we hope you will like this Discovery Piscine.

Chapter II

General instructions

Unless explicitly specified, the following rules will apply to this project.

- This subject is the one and only trustable source. Don't trust any rumor.
- Your assignments will be evaluated by your peers.
- You have a question? Ask your left neighbor. Otherwise, try your luck with your right neighbor.
- You must read the examples thoroughly. They can reveal requirements that are not obvious in the assignment's description.
- Good luck and good work on this project!

Chapter III

Preamble

This project is about Expert Systems. Expert systems are small pieces of logic with very simple tasks. They usually represent state machines, meaning that the machine can only be in one specific state at a time, and some external events make the state change.

Here is what our friend chatGPT has to tell about expert systems:
An Expert System is a basic type of Artificial Intelligence that uses a set of predefined rules and knowledge to simulate the decision-making ability of a human expert in a specific field. It relies on a knowledge base of facts and rules, and an inference engine to apply these rules to given inputs, providing logical conclusions or recommendations. Unlike more advanced AI, Expert Systems don't learn from data but operate solely within the boundaries of their programmed knowledge.

In this project, we will use a very simple and classic expert system: your air conditioning in your house, which tries to keep the indoor temperature withing the comfort zone when the temperature outside changes.

Chapter IV

Exercise 0: Environment

	Exercise : 00
Exercise 0: Environment	
Turn-in directory : $ex00/$	
Files to turn in : `environment.py`	
Forbidden functions : `None`	

We first need to create the extern environment: the outside temperature of your house. Regulation of your indoor temperature will be achieved in the second exercise.

Create a Python program that simulates and prints the daily temperature variations. A scaling factor is applied to simulate a full day in less than a minute. This simulation will represent the environment for the expert system in the next exercise. The program will also provide a different simulation for each season.

`Requirements`:

- A mandatory input parameter specifies the season (it can be a number or the name of the season).
- Each second, the program displays a floating number representing the temperature for a slice of the day (e.g., 30 minute slices, so a full day is covered in 48 seconds).
- Small random variations may be introduced from one cycle to the next.
- The choice of season must affect the simulated temperatures.

You can choose any simulation approach you prefer, as long as it is somewhat realistic. It could be sine-based, linear with multiple segments, or something else.

`Possible execution`:

```
$ python environment.py spring
7.5
8.0
8.5
9.0
10.0
11.0
12.0
13.25
14.5
15.0
15.5
15.5
[...]
```

Chapter V

Exercise 1: Expert System

	Exercise : 01
Exercise 1: Expert System	
Turn-in directory : $ex01/$	
Files to turn in : `expert_system.py`	
Forbidden functions : `None`	

We will now use an expert system to ensure a thermal comfort inside the house.

Create a Python program that reads the temperature from standard input and chooses whether to activate or deactivate heating and cooling.
The program takes command line parameters to set thresholds that trigger heating and cooling.
The indoor temperature is increased or decreased when the air conditioning is on.
Finally, the program outputs the external temperature (received as input), the current action, and the indoor temperature.

`Requirements`:

- The program accepts as many parameters as needed (ex: min-max temperatures outside, indoor temperature, thresholds, etc.).
- For each input received, the program displays on one line the external temperature, the current action of the air conditioning (e.g., heating, cooling, nothing), and the indoor temperature.
- At the very beginning, the indoor temperature is the same as the outside temperature.
- When the heating is turned on, your indoor temperature should raise. A possible strategy would be: start to add 0.5 degrees to the outside temperature. For the next input (1 sec later in the real world, or 30 min later in our simulation), add another 0.5, for a total of 1.0 degree, and so on. Use the same approach for the cooling system, but in the opposite direction, of course.

- When heating or cooling is turned off, each 1 sec / 30 min slice should gradually show that the indoor temperature moving toward the outside temperature. A possible strategy is to move by 0.25 degrees in the direction of the outside temperature.

As you can see, you have to define your own strategy to maintain the indoor temperature as close as possible to a chosen comfort zone.

`Possible execution`:

```
$ python environment.py spring | python expert\_system.py 13.0 15.0
7.5 - heating - 8.0
8.0 - heating - 9.0
8.5 - heating - 10.0
9.0 - heating - 11.0
10.0 - heating - 12.5
11.0 - heating - 14.0
12.0 - nothing - 13.75
13.25 - nothing - 13.5
14.5 - nothing - 13.75
15.0 - nothing - 14.0
15.5 - nothing - 14.25
15.5 - nothing - 14.5
15.5 - nothing - 14.75
16.0 - nothing - 15.0
16.0 - cooling - 14.5
15.5 - cooling - 13.5
15.5 - nothing - 13.75
[...]
```

Your results may vary, as you can decide whether the temperature inside the house is taken at the beginning or at the end of the slice, whether the 0.25 shift toward the outside temperature is always applied, and so on. Your personal algorithm with its own behaviors will certainly impact the result.

Note: Our expert system is not triggered in the middle of a slice, only when it receives a new input. It's not realistic but definitely simpler.

Chapter VI

Submission and peer-evaluation

- Create an `ai-ml_discovery` folder at the root of your home, and move around in it.

- Create a new module folder with this format:

```
module[number_of_the_module]
example: module3 (for module 3)
```

  and navigate to it.

- From now on, all exercises should be in the correct folder rendering (exercise of module 0, in the module0 folder, exercise of module 1, in the module1 folder, etc.). for example:

```
[ai-ml_discovery/module0/[exercise_name].py]
[ai-ml_discovery/module1/[exercise_name].py]
etc...
```



Please note, during your defense anything that is not present in the folder for the day will not be checked.